

Machine Learning and AI in Finance

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PROFESSOR: Fabrizio Ghezzi

EMAIL: fghezzi@ucsd.edu

OFFICE HOURS: by appointment. Contact the instructor via email.

TEACHING ASSISTANT/TUTOR: TBA

EMAIL: TBA

DESCRIPTION

This course introduces students to cutting-edge Machine Learning and AI methodologies in financial applications. Beginning with an overview of alternative data, such as text data, students will explore techniques for analyzing these data and will be exposed to the latest methodologies in the field. The course emphasizes how such data and methods can improve financial decision-making, particularly in shaping portfolio choices, systematic strategies, and trading signals.

OBJECTIVES

The course aims to provide students with an overview of the latest advances in Machine Learning and AI tools used for analyzing financial markets and making financial decisions. Additionally, it equips students with a toolkit to analyze these datasets and stay up to date with the latest research as it becomes available. The ultimate goal of the course is to give students insights on how alternative data and AI tools can be used by industry players — such as asset managers, investors, trading firms — to improve their decision-making process. As such, the course will emphasize practical financial applications through hands-on programming and data analysis. Guest speakers from industry will share practical perspectives on how research developments are being applied in real-world financial settings. Finally, students will have the opportunity to apply the techniques covered during the course of their group project.

MATERIALS

Main Material:

- **Proprietary Course Material:** Slides, Notes, Coding Programs.

Recommended Material:

- **Deep Learning with Python**, by F. Chollet;
- **Deep Learning**, by Goodfellow, Bengio and Courville;

STRUCTURE AND TOPICS

We begin by reviewing new data sources and learning how they are mapped into the statistical domain. Next, we learn to apply the latest econometric and machine learning tools to extract statistical information and patterns that serve as key inputs for financial decision-making. The emphasis is on the underlying financial problems and how these are shaped by the latest advancements in AI. Finally, we investigate issues that may arise from an incorrect use of the methodologies we learn during the course.

SCHEDULE

1. Introduction to alternative and unstructured data sources;
2. Structure and modeling of alternative data;
3. AI methodologies to analyze alternative data;
4. Review of the latest research in finance;
5. Financial applications.

GRADING AND EVALUATION

- **Group Project (50%)** : students are required to design, execute, and craft a research idea which is rooted around the topics of the course. The project must be done in group, and the purpose is to mimic how quant research pods work in quantitative investment firms.
- **Project Presentation (25%)**: students will present (pitch) their research idea to the classroom. The object should be to convince the audience of the quality of their idea.
- **Class Participation (25%)**: actively engage and following classes.

SUMMARY OF GRADING

Assignments	Points
Group Project	50
Presentation of Group Project	25
Class Participation	25
Total	100

COURSE POLICIES

[Include whatever course policies haven't been covered in the rest of the syllabus; these might include information about attendance, class participation, and making up work.]

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.